

Q1.

1. Main Features, Uses, and Various Versions of HTML

HTML (HyperText Markup Language) is the standard language used to create web pages. It provides the basic structure of a website and works together with CSS and JavaScript.

Features of HTML

- Easy to learn and use.
- Platform independent and works on all browsers.
- Uses tags to structure content.
- Supports links, images, tables, forms, and multimedia.
- Can be integrated with CSS and JavaScript.

Uses of HTML

- Creating web pages and websites.
- Displaying text, images, videos, and audio.
- Creating forms for user input.
- Linking different web pages together.

Versions of HTML

- **HTML 1.0** – Basic version with simple text support.
- **HTML 2.0** – Added forms and improved features.
- **HTML 3.2** – Added tables and scripting support.
- **HTML 4.01** – Better styling and multimedia support.
- **XHTML** – Stricter version of HTML.
- **HTML5** – Latest version with audio, video, graphics, and mobile-friendly features.

2. Basic Structure of an HTML Document

Every HTML document follows a standard structure.

```
<!DOCTYPE    html>
<html>
<head>
  <title>My      Web      Page</title>
</head>
<body>
  <h1>Welcome to      My      Website</h1>
  <p>This      is      a      sample  webpage.</p>
</body>
</html>
```

Head Section

The <head> section contains information about the webpage that is not directly displayed on the screen.

It includes:

- Page title (<title>)
- Meta tags
- Links to CSS files
- JavaScript files

Example:

```
<head>
  <title>College Website</title>
</head>
```

Body Section

The <body> section contains all the visible content of the webpage.

It includes:

- Headings
- Paragraphs
- Images
- Tables
- Forms
- Links

Example:

```
<body>
  <h1>Hello Students</h1>
  <p>Welcome to HTML.</p>
</body>
```

3. Development of HTML for Modern Web Requirements

HTML has changed significantly over time to meet modern web development needs. Earlier versions mainly focused on displaying text and links. As websites became more advanced, new features were added.

The introduction of **HTML5** brought many improvements such as:

- Native support for audio and video.
- Better form controls.
- Canvas element for graphics.
- Improved performance and multimedia support.

Support for Mobile Devices

HTML5 supports mobile devices through the **viewport meta tag**, which helps webpages adjust to different screen sizes.

Support for Responsive Design

HTML works with CSS media queries and flexible layouts to create responsive websites. Responsive design allows a webpage to automatically adapt to desktops, tablets, and smartphones, providing a better user experience.

Q2.

1. Definition of CSS and Types of CSS

CSS (**Cascading Style Sheets**) is a style sheet language used to control the appearance and layout of HTML webpages. It helps make websites attractive and user-friendly.

Types of CSS

1. Inline CSS

```
<p style="color:blue;">Hello</p>
```

Style is applied directly to an HTML element.

2. Internal CSS

```
<style>
p{color:blue;}
</style>
```

Style is written inside the <head> section.

3. External CSS

```
<link rel="stylesheet" href="style.css">
```

Styles are stored in a separate CSS file and linked to the webpage.

2. Role of CSS in Web Development

CSS plays an important role in website design by:

- Improving webpage layout and appearance.
- Maintaining a consistent design across multiple pages.
- Reducing code repetition.
- Making websites responsive for mobiles, tablets, and desktops.
- Enhancing user experience through better styling.

3. Classes, IDs, Selectors, and Common CSS Issues

Classes

Classes are used to apply the same style to multiple elements.

```
<p class="text">Hello</p>
```

IDs

IDs are used to style a unique element.

```
<p id="heading">Welcome</p>
```

Selectors

Selectors help target HTML elements for styling.

Examples:

```
p{color: red;}  
.text{color:blue;}  
#heading{ color: green; }
```

Q3.

1. HTML Tables and Their Uses

HTML tables are used to organize and display data in rows and columns.

Common table tags:

- <table> – Creates a table
- <tr> – Creates a row
- <th> – Creates a heading cell
- <td> – Creates a data cell

Example:

```
<table border="1">  
<tr>  
<th>Name</th>  
<th>Age</th>  
</tr>  
<tr>  
<td>Rahul</td>  
<td>20</td>  
</tr>  
</table>
```

Tables help present information in a clear and organized manner, making data easier to read and compare.

2. Lists and Text Formatting Tags

Types of Lists

1. Ordered List ()

- Displays items with numbers.

```
<ol>  
<li>HTML</li>  
<li>CSS</li>  
</ol>
```

2. Unordered List ()

- Displays items with bullets.

```
<ul>  
<li>HTML</li>  
<li>CSS</li>  
</ul>
```

3. Description List (<dl>)

- Used for terms and definitions.

```
<dl>  
<dt>HTML</dt>  
<dd>HyperText Markup Language</dd>  
</dl>
```

Text Formatting Tags

- – Bold text
- <i> – Italic text
- <u> – Underlined text
- – Important text
- – Emphasized text

These tags improve readability and make important content stand out.

3. Accessibility and Semantic Best Practices

To ensure proper accessibility and semantic usage:

- Use <th> for table headings instead of normal cells.
- Add meaningful headings and labels.
- Use tables only for data, not for page layout.
- Use , , and <dl> according to the type of content.
- Keep list items inside tags.

- Use semantic tags like and instead of only visual tags.
- Maintain a clear structure so screen readers can understand the content.

Q4.

1. Structure of HTML Forms, Input Types, and Attributes

HTML forms are used to collect information from users and send it to a server for processing.

Basic form structure:

```
<form action="submit.php" method="post">
<label>Name:</label>
<input type="text" name="username">
<input type="submit" value="Submit">
</form>
```

Common Input Types

- text – Text input
- password – Password field
- email – Email input
- radio – Single choice option
- checkbox – Multiple choices
- submit – Submit button

Important Attributes

- name – Identifies the input field
- value – Default value
- placeholder – Displays hint text
- required – Makes field mandatory
- maxlength – Sets character limit

2. Interactivity in Web Applications

Forms make websites interactive by allowing users to:

- Enter personal information.
- Register and log in.
- Submit feedback and surveys.
- Search for information.
- Upload files and send requests.

Without forms, users would only be able to view content and not interact with the website.

3. Form Validation, User Experience, and Accessibility

Form Validation Techniques

Form validation ensures that users enter correct data.

- Use required to prevent empty fields.
- Use type="email" for valid email addresses.
- Use maxlength to limit input length.
- Use JavaScript for advanced validation.

Enhancing User Experience

- Provide clear labels and instructions.
- Use placeholders for guidance.
- Display error messages clearly.
- Group related fields together.
- Keep forms simple and easy to fill.

Improving Accessibility

- Use <label> tags for all input fields.
- Ensure proper keyboard navigation.
- Use meaningful field names.
- Provide clear error messages.
- Maintain good color contrast and readable text.

Q5.

1. Adding and Aligning Text in HTML

Text can be added in HTML using different tags.

Common Text Tags

- <h1> to <h6> – Headings
- <p> – Paragraph
- – Bold text
- <i> – Italic text
- <u> – Underlined text

Example:

```
<h1>Welcome</h1>  
<p>This is a paragraph.</p>
```

Text Alignment

Text alignment can be done using CSS.

```
<p style="text-align:center;">Centered Text</p>
```

Types of alignment:

- Left
- Right
- Center
- Justify

Proper text formatting improves readability and presentation.

2. Adding Images and Hyperlinks in HTML

Images

Images are added using the `<img>` tag.

```

```

Important attributes:

- `src` – Image location
- `alt` – Alternative text
- `width` and `height` – Image size

Hyperlinks

Hyperlinks are created using the `<a>` tag.

```
<a href="https://example.com">Visit Website</a>
```

The `href` attribute specifies the destination of the link.

3. Image Optimization and Accessible Links

Best Practices for Image Optimization

- Use compressed images to reduce loading time.
- Choose suitable formats such as JPG, PNG, or WebP.
- Specify image dimensions.
- Use meaningful file names.
- Add alt text for accessibility.

Best Practices for Hyperlinks

- Use clear and descriptive link text.
- Avoid vague text like "Click Here".
- Ensure links work correctly.
- Make links easy to identify.
- Use relevant keywords for better SEO.

Accessibility and SEO Benefits

- Alt text helps screen readers understand images.

- Descriptive links improve navigation for users.
- Search engines can better understand webpage content.

Q6.

1. JavaScript Syntax Rules, Data Types, and Operators

JavaScript is a scripting language used to make web pages interactive.

Basic Syntax Rules

- Statements end with a semicolon (;).
- Variables are declared using var, let, or const.
- JavaScript is case-sensitive.
- Code is written inside <script> tags.

Common Data Types

- **Number** – 10, 25.5
- **String** – "Hello"
- **Boolean** – true, false
- **Array** – [1, 2, 3]
- **Object** – {name:"Rahul"}

Common Operators

- Arithmetic: +, -, *, /
- Comparison: ==, !=, >, <
- Logical: &&, ||, !

2. Concept of JSON

JSON (**JavaScript Object Notation**) is a lightweight format used to store and exchange data between applications.

Example:

```
{  
  "name":"Rahul","age":20  
}
```

Features of JSON:

- Easy to read and write.
- Lightweight and fast.
- Language-independent.
- Widely used in web applications.

JSON represents data as key-value pairs.

3. JSON Parsing, String Conversion, and Applications

Converting JSON String to Object

JSON.parse() is used to convert a JSON string into a JavaScript object.

```
let obj = JSON.parse('{"name":"Rahul"}');
```

Converting Object to JSON String

JSON.stringify() is used to convert a JavaScript object into a JSON string.

```
let str = JSON.stringify({name:"Rahul"});
```

Applications of JSON

- Data exchange between client and server.
- Web APIs and REST services.
- Storing configuration data.
- Sending and receiving data in AJAX requests.
- Communication between web and mobile applications.